

Week 13 Discussion Solutions

1. Consider a long solenoid with turn density n , radius R , length ℓ , and current I .

- (a) For a long solenoid, choose an Amperian rectangle with one long side inside the solenoid and one outside. Neglecting fringing fields,

$$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = BL = \mu_0 I_{\text{enc}} = \mu_0(nL)I,$$

so

$$\boxed{B = \mu_0 n I}$$

inside the solenoid, directed along the axis by the right-hand rule.

- (b) A single circular loop produces on its axis

$$B_{\text{loop}}(z) = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}.$$

A solenoid can be treated as a stack of loops with turn density n , so

$$B = \int_{-\infty}^{\infty} n dz \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}.$$

Let $z = Rx$, so $dz = Rdx$. Then

$$B = \frac{\mu_0 n I}{2} \int_{-\infty}^{\infty} \frac{dx}{(1 + x^2)^{3/2}} = \frac{\mu_0 n I}{2} (2) = \boxed{\mu_0 n I}.$$

This agrees with part (a).

- (c) The magnetic flux through one turn is

$$\Phi_B = BA = (\mu_0 n I)(\pi R^2).$$

The number of turns is $N = n\ell$, so the total flux linkage is

$$N\Phi_B = (n\ell)(\mu_0 n I \pi R^2).$$

Hence

$$L = \frac{N\Phi_B}{I} = \boxed{\mu_0 n^2 \pi R^2 \ell}.$$

Equivalently, using $N = n\ell$,

$$\boxed{L = \mu_0 \frac{N^2 \pi R^2}{\ell}}.$$

2. A toroidal coil of inner radius r_1 , outer radius r_2 , and height h contains N turns and carries a current I .

- (a) By symmetry, the magnetic field is tangent to circles centered on the toroid axis. Using Ampère's law on a circular path of radius r with $r_1 < r < r_2$,

$$\oint \mathbf{B} \cdot d\ell = B(2\pi r) = \mu_0 N I.$$

Therefore

$$B(r) = \frac{\mu_0 N I}{2\pi r}, \quad r_1 < r < r_2.$$

Outside the windings, $B \approx 0$ for an ideal toroid. The direction follows the right-hand rule around the torus.

- (b) The flux through one turn is obtained from the cross-sectional area element $dA = h dr$:

$$\Phi_B = \int_{r_1}^{r_2} B(r) dA = \int_{r_1}^{r_2} \frac{\mu_0 N I}{2\pi r} h dr = \frac{\mu_0 N I h}{2\pi} \ln \frac{r_2}{r_1}.$$

Thus the inductance is

$$L = \frac{N \Phi_B}{I} = \frac{\mu_0 N^2 h}{2\pi} \ln \frac{r_2}{r_1}.$$

- (c) The magnetic energy is

$$U = \int \frac{B^2}{2\mu_0} d\tau.$$

Here $d\tau = (2\pi r)(h dr)$, so

$$U = \int_{r_1}^{r_2} \frac{1}{2\mu_0} \left(\frac{\mu_0 N I}{2\pi r} \right)^2 (2\pi r h dr) = \frac{\mu_0 N^2 I^2 h}{4\pi} \int_{r_1}^{r_2} \frac{dr}{r}.$$

Hence

$$U = \frac{\mu_0 N^2 I^2 h}{4\pi} \ln \frac{r_2}{r_1}.$$

- (d) From $U = \frac{1}{2} L I^2$,

$$L = \frac{2U}{I^2} = \frac{\mu_0 N^2 h}{2\pi} \ln \frac{r_2}{r_1},$$

which matches part (b).

3. Consider two long coaxial solenoids with turn densities n_1, n_2 , cross-sectional areas A_1, A_2 , and common length h .

- (a) Inside solenoid 1,

$$B_1 = \mu_0 n_1 I_1$$

for an ideal long solenoid.

- (b) The field from solenoid 1 is essentially uniform over the overlap region. The flux through one turn of solenoid 2 is therefore

$$\Phi_{21} = B_1 A_{\text{overlap}} = \mu_0 n_1 I_1 A_{\text{overlap}},$$

where

$$A_{\text{overlap}} = \min(A_1, A_2) = A_1$$

for coaxial nested solenoids.

- (c) Solenoid 2 has $N_2 = n_2 h$ turns, so the total linked flux is

$$N_2 \Phi_{21} = n_2 h \mu_0 n_1 I_1 A_1.$$

Thus

$$M = \frac{N_2 \Phi_{21}}{I_1} = \boxed{\mu_0 n_1 n_2 A_1 h}.$$

- (d) No. Mutual inductance is symmetric:

$$\boxed{M_{21} = M_{12}}.$$

So it does not depend on which solenoid you call “1” or “2.”

4. A non-conducting circular disk of negligible thickness and radius R carries a uniformly distributed charge Q and spins with angular speed ω .

- (a) The surface charge density is

$$\sigma = \frac{Q}{\pi R^2}.$$

A thin ring of radius r and width dr carries charge

$$dq = \sigma(2\pi r dr).$$

Since the period is $T = 2\pi/\omega$, the ring current is

$$dI = \frac{dq}{T} = \sigma \omega r dr.$$

Its area is πr^2 , so the infinitesimal magnetic dipole moment is

$$d\mu = dI(\pi r^2) = \boxed{\pi \sigma \omega r^3 dr}.$$

Its direction is along the symmetry axis by the right-hand rule.

- (b) Integrating from 0 to R ,

$$\mu = \int_0^R \pi \sigma \omega r^3 dr = \pi \sigma \omega \frac{R^4}{4}.$$

Using $\sigma = Q/(\pi R^2)$ gives

$$\boxed{\mu = \frac{Q \omega R^2}{4}}.$$

In vector form, μ points along the rotation axis.

(c) The torque in an external field is

$$\tau = \mu \times \mathbf{B},$$

so its magnitude is

$$\tau = \mu B \sin \theta = \frac{Q\omega R^2}{4} B \sin \theta.$$

5. Consider a resistor R , capacitor C , inductor L , and battery with emf $\mathcal{E}$ all in series.

(a) Kirchhoff's loop law gives

$$\mathcal{E} - L \frac{dI}{dt} - RI - \frac{Q}{C} = 0, \quad I = \frac{dQ}{dt}.$$

Hence

$$L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = \mathcal{E}.$$

(b) The homogeneous equation has characteristic roots

$$r_{\pm} = \frac{-R \pm \sqrt{R^2 - 4L/C}}{2L}.$$

A constant particular solution is $Q_p = C\mathcal{E}$. Therefore

$$Q(t) = C\mathcal{E} + c_1 e^{r_+ t} + c_2 e^{r_- t}.$$

For a physically standard switch-on from rest, the correct initial conditions are

$$Q(0) = 0, \quad I(0) = \dot{Q}(0) = 0,$$

because the inductor current cannot jump instantaneously. Using these,

$$c_1 + c_2 = -C\mathcal{E}, \quad r_+ c_1 + r_- c_2 = 0,$$

so

$$Q(t) = C\mathcal{E} \left[1 + \frac{r_- e^{r_+ t} - r_+ e^{r_- t}}{r_+ - r_-} \right].$$

Also,

$$I(t) = \dot{Q}(t) = C\mathcal{E} \frac{r_+ r_-}{r_+ - r_-} (e^{r_+ t} - e^{r_- t}).$$

Now classify the motion:

- If $R^2 > 4L/C$ (equivalently $RC > 4L/R$), the roots are real and negative: **overdamped**.
- If $R^2 < 4L/C$ (equivalently $RC < 4L/R$), the roots are complex with negative real part: **underdamped** oscillations.
- If $R^2 = 4L/C$, the circuit is **critically damped**.

As $t \rightarrow \infty$, the capacitor charges to

$$Q(\infty) = C\mathcal{E}.$$

6. A conducting bar of length ℓ slides with speed v on frictionless rails in a uniform magnetic field B perpendicular to the loop. The circuit has resistance R .

- (a) If the bar moves so the enclosed area increases at rate

$$\frac{dA}{dt} = \ell v,$$

then the magnetic flux changes at rate

$$|\mathcal{E}_{\text{ind}}| = \left| \frac{d\Phi_B}{dt} \right| = B\ell v.$$

So the induced current magnitude is

$$I = \frac{B\ell v}{R}.$$

The electrical power dissipated is

$$P_{\text{elec}} = I^2 R = \frac{B^2 \ell^2 v^2}{R}.$$

- (b) The magnetic force on the bar has magnitude

$$F_B = I\ell B = \frac{B\ell v}{R} \ell B = \frac{B^2 \ell^2}{R} v,$$

and it opposes the motion (Lenz's law). The mechanical power delivered against this force is

$$P_{\text{mech}} = F_B v = \frac{B^2 \ell^2 v^2}{R},$$

which exactly matches the Joule-heating rate from part (a).

- (c) Since the magnetic force is the only horizontal force,

$$m \frac{dv}{dt} = -\frac{B^2 \ell^2}{R} v.$$

Thus $v(t) = v_0 e^{-t/\tau}$ with

$$\tau = \frac{mR}{B^2 \ell^2}.$$

The total energy dissipated equals the loss of kinetic energy:

$$E_{\text{diss}} = \frac{1}{2} m v_0^2.$$